

EC2 Setup Description

I am using a t2.large EC2 instance with a docker container that runs the execution kernel.

Dataset

The month used from the dataset is January of 2026.

The following data cleaning assumptions were made:

- Removed rows with missing pickup or drop-off timestamps
- Removed trips with non-positive distance
- Removed trips with negative fares
- Removed trips with negative total amount
- Removed trips longer than 24 hours
- Removed trips with drop-off time before pickup time
- Added taxi_type column
- For the regression model, taxi_type was encoded into binary integer
- For the regression model, any rows with null values were dropped

Responses to Analytics Questions

1. The yellow taxi type had more trips than the green taxi type, almost 100 times more.

```
+-----+-----+
| taxi_type | count |
+-----+-----+
|   yellow | 3562033 |
|   green  |   38913 |
+-----+-----+
```

2. The average fare for yellow taxis was \$21.02 and \$16.00 for green taxis.

```
+-----+-----+
| taxi_type | avg(fare_amount) |
+-----+-----+
|   yellow | 21.021527776420072 |
|   green  | 16.004368719965075 |
+-----+-----+
```

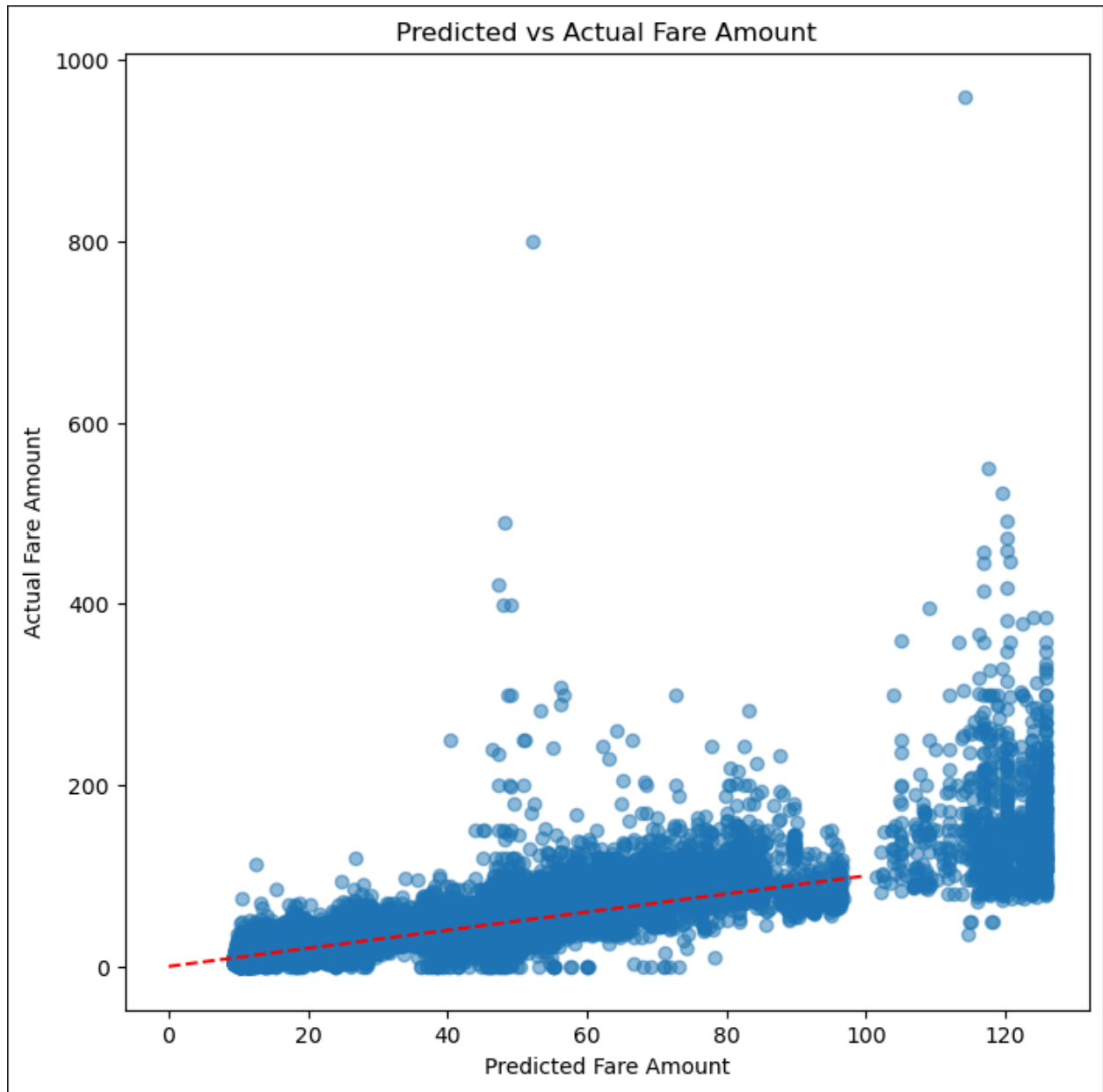
3. The average trip distance for yellow taxis was 6.7 miles and 13.0 miles for green taxis.

```
+-----+-----+
| taxi_type | avg(trip_distance) |
+-----+-----+
|   yellow | 6.705125379241041 |
|   green  | 12.988674221982372 |
+-----+-----+
```

4. The 18th hour of the day had the most pickups.

pickup_hour	count
18	235765
17	227332
15	219306
19	213794
20	210273
21	209653
16	209371
14	205223
13	193716
22	192096
12	185623
11	170266
10	160406
9	156308
23	150074
8	148023
7	113377
0	109932
1	76311
6	63194

8. The training RMSE was 6.65 and the test RMSE was 6.51. For my random forest regression model, I used all numerical features except those that were directly included in the total fee and those which had most of their entries missing. Out of the features that were used, the most important seem to be trip_distance and RatecodeID since those are most likely used as a baseline for calculating the fare amount (distance X rate = fare). Following that feature, PULocationID, DOLocationID, and taxi_type are good supporting predictors because some areas may affect the price, and we know from earlier analysis that the green and yellow taxis have different average fare costs. Finally, VendorID, passenger_count, and payment_type, may also be helpful supporting predictors since different vendors may have different patterns, and the amount of passengers and whether a customer uses cash or card can be loosely correlated with how long of a trip a customer requests.



S3 Output Paths

Question 1 trips by taxi type: s3://rosu-de300spring2026-bucket/HW3/trips_by_type

Question 2 average fare by taxi type: s3://rosu-de300spring2026-bucket/HW3/avg_fare_by_type

Question 3 average distance by taxi type:

s3://rosu-de300spring2026-bucket/HW3/avg_distance_by_type

Question 4 pickups for each hour: s3://rosu-de300spring2026-bucket/HW3/pickups_by_hour

Question 8 predicted vs actual fares:

s3://rosu-de300spring2026-bucket/HW3/predicted_vs_actual_fare.png